Yael Kirkpatrick

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Homepage

Research Interests

Graph Algorithms, Fine Grained Complexity, Approximation Algorithms

Academic Background

Ph.D. Applied Mathematics 2021 - Exp. 2027
Massachusetts Institute of Technology
Advisor: Virginia Vassilevska Williams
GPA: 5.0/5.0

B.S. Mathematics and Computer Science 2019 - 2021
Hebrew University of Jerusalem, Israel
Summa Cum Laude with highest departmental GPA
GPA: 98 / 100

Publications

14. Yael Kirkpatrick. "Improved Approximation of Min-Distances in Near-Linear Time." SODA 2027. **Best Student Paper Award.**
13. Sam Hiken, Yael Kirkpatrick, Jakob Nogler and Virginia Vassilevska Williams. "The Cost of Changing Edges for Diameter Computation and More." SODA 2027.
12. Yael Kirkpatrick, John Kuszmaul, Merey Temirzinova, Virginia Vassilevska Williams. "When Shall We k Meet Again? Tight Algorithms for Diameter and Radius under the Meet Distance." ITCS 2027.
11. Yael Kirkpatrick, Liam Roditty, Richard Qi and Virginia Vassilevska Williams. "New Diameter Approximations via Distance Oracle Techniques." ICALP 2026.
10. Yael Kirkpatrick, John Kuszmaul, Surya Mathialagan, Virginia Vassilevska Williams. "Preprocessed 3SUM for Unknown Universes with Subquadratic Space." ICALP 2026.
9. Ce Jin, Yael Kirkpatrick, Michał Stawarz and Virginia Vassilevska Williams. "Improved Additive Approximation Algorithms for APSP." SODA 2026 and HALG 2026.
8. Yael Kirkpatrick and Virginia Vassilevska Williams. "Shortest Paths in Multi-mode Graphs." MFCS 2025.
7. Ce Jin, Yael Kirkpatrick, Virginia Vassilevska Williams and Nicole Wein. "Beyond 2-Approximation for k-Center in Graphs." SODA 2025.
6. Hayashi Ani, Josh Brunner, Erik D. Demaine, Jenny Diomidova, Timothy Gomez, Della Hendrickson, Yael Kirkpatrick, Jeffery Li, Jayson Lynch, Ritam Nag and Frederick Stock. "Agent Motion Planning as Block Asynchronous Cellular Automata: Pushing, Pulling, Suplexing, and More." UCNC 2024.
5. Erik Demaine, Yael Kirkpatrick, and Rebecca Lin. "Graph Threading." ITCS 2024.
4. Michal Dory, Sebastian Forster, Yael Kirkpatrick, Yasamin Nazari, Virginia Vassilevska Williams, and Tijn de Vos. "Fast 2-Approximate All-Pairs Shortest Paths." SODA 2024.

3. Yael Kirkpatrick, Rachel Dudukovich, Prash Choksi, and Dominick Ta. "Cooperative clustering techniques for space network scalability." In 2023 IEEE Cognitive Communications for Aerospace Applications Workshop (CCAAW), IEEE, 2023.
2. Michael Moy, Robert Kassouf-Short, Nadia Kortas, Jacob Cleveland, Brian Tomko, Dominic Conricode, Yael Kirkpatrick, Robert Cardona, Brian Heller, and Justin Curry. "Contact Multigraph Routing: Overview and Implementation." In 2023 IEEE Aerospace Conference, 2023.
1. Mingyang Deng, Yael Kirkpatrick, Victor Rong, Virginia Vassilevska Williams, and Ziqian Zhong. "New additive approximations for shortest paths and cycles." ICALP 2022.

Preprints

3. Erik Demaine, Yael Kirkpatrick, and Rebecca Lin. "Graph Threading with Turn Costs." Preprint, 2024.
2. Yael Kirkpatrick and Surya Mathialagan. "On the Conditional Optimality of the Chiba-Nishizeki Subgraph Enumeration Algorithm." Preprint, 2024.
1. Gal Beniamini, Asaf Etgar and Yael Kirkpatrick. "The F2-Rank and Size of Graphs." Preprint, 2020.

Awards

Best Student Paper Award SODA 2027
Awarded for the solo-authored paper "Improved Approximation of Min-Distances in Near-Linear Time."

National Science Foundation (NSF) Graduate Research Fellow 2023-2027
Awarded to outstanding graduate students who have demonstrated the potential to be high achieving scientists and engineers, early in their careers.

Lightricks Excellence Scholarship Program 2020 - 2021
Awarded annually for academic excellence among students in the Hebrew University's Computer Science department.

Dean's List and Dean's Prize 2019 - 2021
Awarded for academic years 2019-2020 and 2020-2021 by the Computer Science Department of the Hebrew University.

Research Experience

Graduate Researcher, MIT Fall 2021-Present
Advisor: Prof. Virginia Vassilevska Williams.

Research Intern, NASA June 2022-Sept 2022
Supervisor: Dr. Rachel Dudukovich, NASA Glenn Research Center, Cleveland OH.
Research on clustering techniques applied to contact graph routing.

Research Assistant, Hebrew University July 2020-May 2021
Supervisor: Prof. Israel Nelken, The Laboratory of Auditory Neurophysiology, ELSC.
Research in the geometry of dimensionality reduction of neural data.

Researcher (Military), 8200 Unit, IDF March 2016 - Sept 2019
Technology Center of 8200 Intelligence Unit of the IDF.
Signal analysis and communication systems research.

Teaching Experience

Glassblowing Instructor, MIT Fall 2025 - Fall 2026 (4 sessions)
Instructor in the beginner class of the MIT glass lab.

Teaching Assistant, MIT Spring 2025
Teaching assistant for MIT course 6.5350, Matrix Multiplication and Graph Algorithms. The course explores graduate topics around matrix multiplication and its use in the design of graph algorithms.

Recitation Instructor, MIT Fall 2024
Recitation instruction for MIT course 18.01, Single Variable Calculus. The course covers introductory calculus including differentiation and integration of functions of one variable, differential equations and Taylor series, with applications.

Instructor, PRIMES TCS Circle Spring + Fall 2024
Teacher in MIT PRIMES circle in theoretical computer science, giving weekly lectures to high school students on topics in TCS, spanning algorithms, cryptography, data structures, machine learning and more.

Teaching Assistant, Hebrew University Spring 2021
Teaching assistant for the Hebrew University course 80316, Advanced Infinitesimal Calculus II. The course covers analysis of manifolds and multivariate calculus.

Guest Lecturer, Israeli Young Mathematics Olympiad Team 2020-2022
I gave various lectures as part of the training of the Israeli Young Mathematics Olympiad Team, covering subjects in geometry, graph theory and spacial transformations.

Teacher, 8200 School of Cyber 2016-2019
I taught various classes at the 8200 intelligence unit school of cyber. Subjects include signal analysis, communication systems and protocols, information theory, infinitesimal calculus, linear algebra and probability.

Talks

Improved Approximation of Min-Distances in Near-Linear Time, Mind-IL Poster Session, 2026.

New Linear-Time Algorithms for Min-Distance Approximations, Weizmann Algorithms Seminar, 2026.

Harnessing Matrix Multiplication for Additive APSP Approximation, DIMACS Workshop on Algebraic Techniques in Fine-Grained Complexity, 2026.

Color Fault-Tolerant Distance Preservers: Optimal Size in Conditionally Optimal Time, ICALP 2026.

New Diameter Approximations via Distance Oracle Techniques, ICALP 2026.

Shortest Paths in Multimode Graphs, Gdansk Conference on Graph Theory, 2026.

Preprocessed 3SUM for Unknown Universes, Weizmann Algorithms Seminar, 2026.

Harnessing Matrix Multiplication for Additive APSP Approximation, Tel Aviv University Algorithms Seminar, 2026.

Improved Additive Approximation Algorithms for APSP, SODA, 2026.

Harnessing Matrix Multiplication for Additive APSP Approximation, Chicago Junior Theorist Workshop, 2025.

Shortest Paths in Multimode Graphs, MFCS, 2025.

Beyond 2-Approximation for k-Center in Graphs, DIMACS Workshop on Hardness of Approximation in P, 2025. [Recording]

Beyond 2-Approximation for k-Center in Graphs, EnCORE Workshop on Fine Grained Complexity, 2025.

Beyond 2-Approximation for k-Center in Graphs, Tel Aviv Univeristy Algorithms Seminar, 2025.

Beyond 2-Approximation for k-Center in Graphs, SODA, 2025.

On Shortest Paths in Multimode Graphs, Bar Ilan University, 2024.

On Shortest Paths in Multimode Graphs, Weizmann Institute, 2024.

On Shortest Paths in Multimode Graphs, Paris Lodron University of Salzburg, 2024.

Fast 2-Approximate All-Pairs Shortest Paths, SODA, 2024.

Graph Threading, MIT Pure Math Graduate Student Seminar, 2023.

Cooperative Clustering Techniques for Space Network Scalability, IEEE CCAAW, 2023.

New Additive Aprroximations for Shortest Paths and Cycles, ICALP, 2022.

Volunteer Experience

Jewish Big Brothers Big Sisters of Greater Boston 2022 - 2024
I participate in the Jewish Big Brothers Big Sisters of Greater Boston's Friend to Friend program. The program matches volunteers with adults with mild to moderate disabilities for periodic activities, giving them the opportunity to make new friends, socialize, stay active and be connected to the community.

Care of Hospitalized Children 2020 - 2021
I volunteered with Ruach Tova, taking care of hospitalized children and elderly patients while their caregivers were unable to be present.

Volunteering with Holocaust Survivors 2017 - 2020
I volunteered with LaHaim for three years, providing companionship to Holocaust survivors in the community.

Emergency Medical Technician (EMT) 2014 - 2015
I volunteered as a junior EMT on ambulances as part of Israel's national rescue and emergency service organization, Magen David Adom (MDA).